NLME Objective Function

Richard C. Brundage, PharmD, PhD, FISoP, FCP

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University of Minnesota
Driven to Discover®

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Objectives

- Start scratching away at the mystery of the NLME FOCE objective function

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Model = Equation = Function

- A MODEL → equation that produces a family of lines
  - y = mx + b
  - Cp = D/V exp (-(CL/V) * t)
- Dependent Variables
- Independent Variables
- PARAMETERS: m, b, V, CL
  - we often call them what they are: slope, intercept, volume, clearance
  - impose a unique shape

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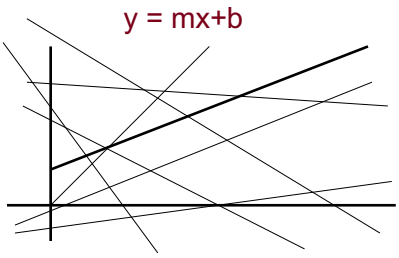
y = mx+b



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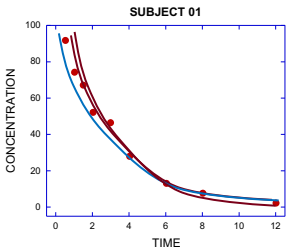
We start with data

Then we propose a MODEL that might reasonably predict the data

Then we need to think about which parameters to choose to best match the data



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How to do that

- Data
- Model
- Parameters
- What does it mean to have the prediction line reasonably go through the data?
- Make the line go as close to as many of the data points as possible

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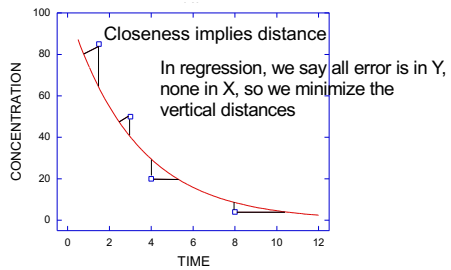
Start by getting qualitative



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Time to get quantitative

$(\text{Cobs} - \text{Cpred})$ Residuals

$\sum(\text{Cobs} - \text{Cpred})$ Sum them up

Problematic: positive and negative

$\sum(\text{Cobs} - \text{Cpred})^2$ SSQ

where

Cobs are the data

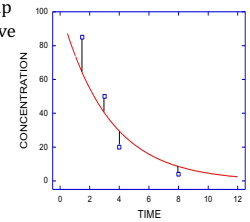
Cpred = D/V * $\exp^{(-kt)}$

D is dose

t is time

k is a first-order rate constant

V is volume of distribution



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Vertical Distances

- Observed data
- Model

  Cp=D/V $\exp^{(-Kt)}$
- Model parameters

  V=48L

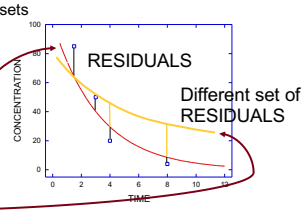
  K=0.412 hr^{-1}

Predicted data

V=42L

K=0.356 hr^{-1}

These residuals now give us a way to compare the closeness of two parameter sets



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Nonlinear Regression Algorithm

1. You must give initial starting values for each parameter in your model
2. Computer calculates the SSQ
3. Search for another parameter set (i.e., new pair of CL, V)
4. Compute new SSQ
5. If new SSQ is lower; loop back to 3, or
6. Stop when a lower SSQ cannot be found

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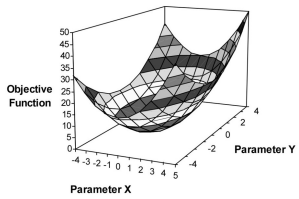
Some Parameter Search (Optimization)

- Steepest descent
  - Iterative gradient (derivative) method
  - Useful when far away from the minimum
- Gauss Newton
  - Approximates surface and solves for minimum
  - Useful when close to minimum
- Marquardt
  - Iterative method that employs steepest descent initially and Gauss Newton when gradient is shallow (near minimum)
- Simplex
  - Fast, iterative, derivative free vertex method
  - No standard errors of the estimate



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Not quite that easy

- Usually more than just 2 parameters
- Consider
  - CL, V, KA
  - Variability on CL
  - Variability on V
  - Variability on KA
  - Residual Unexplained Variability
  - Genotype effect on CL
  - Weight on V

10-dimensional hyperspace

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Objective Function Value

$$OFV_{OLS} = \sum_j (C_{ij,obs} - C_{ij,pred})^2$$

where

in individual i, at time j

$$C_{ij,pred} = \frac{D_i}{V_i} \exp^{(-CL_i / V_i \times t_{ij})}$$

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Analytical Variability

- With most analytical systems, the noise of the response increases in proportion to the signal
- Constant S/N ratio
- Constant Coefficient of Variation (CV)

  CV = SD/mean
- Analytical variability is usually 5-10%; acceptable up to 20% (FDA)

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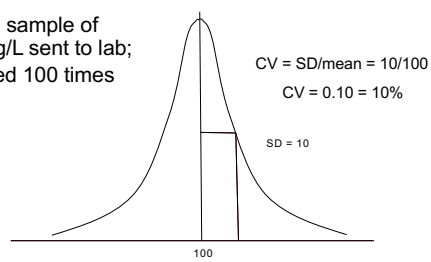
Known sample of
100 mg/L sent to lab;
Assayed 100 times



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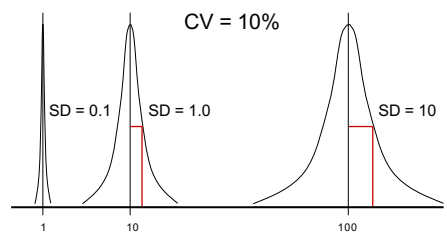

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ZIDOVUDINE
QCs

Sample	75	300	1500
1	75.3	306	1567
2	78.2	299	1553
3	72.6	305	1532
4	69.1	302	1525
5	69.4	287	1510
mean	72.9	300	1537
SD	3.89	7.66	22.66
CV%	**5.30%**	**2.60%**	**1.50%**

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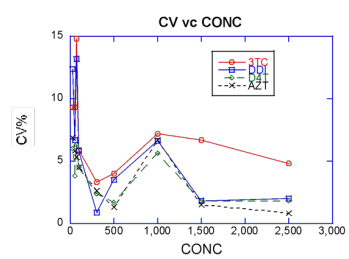

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So why is all this important?

- An assumption of nonlinear regression is that the variance at all observations is equal
- Homoschedasticity
- Clearly not what we have in PK systems

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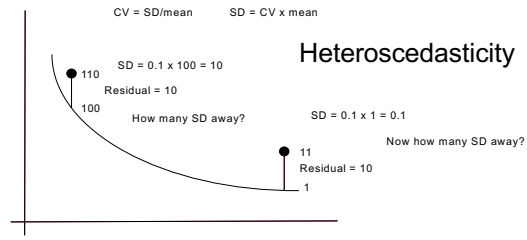
Relationship between SD and residuals



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Objective Function Value (WLS)

$$OFV_{WLS} = \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{W_{ij}}$$

Residuals (RES) are now Weighted Residuals (WRES)

where

in individual i, at time j

$$C_{ij,pred} = \frac{D_i}{V_i} \exp^{(-CL_i/V_i \times t_{ij})}$$

$$W_{ij} = \text{weight} = \text{variance} = SD^2 = (CV \times Cp)^2$$

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$$OFV_{WLS} = \sum_i \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{W_{ij}}$$

where

$$C_{ij,pred} = \frac{D_i}{V_i} \exp^{-CL_i/V_i \times t_{ij}}$$

where

$$CL_i = TVCL_i + ETA_{CL,i}$$

$$V_i = TVV_i + ETA_{V,i}$$

where

$$TVCL_i = THETA(1) + THETA(2) \times CLCR_i$$

$$TVV_i = THETA(3) * WT$$

$$W_{ij} = \text{variance } C_{ij}$$

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Mathematical Difficulties

$$OFV_{WLS} = \sum_j \frac{\left(C_{ij,obs} - \frac{D_i}{V_i} \exp^{-CL_i/V_i \times t_{ij}}\right)^2}{W_{ij}}$$

The OFV can't be solved when ETAs enter the model nonlinearly

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First-Order Approximation

- Using the First-order Taylor Series Expansion
- And set ETAs and EPSILONs to their expected value of zero

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$$\frac{\left(C_{ij,obs} - \frac{D_i}{TVV_i + ETA_V} \exp^{-(TVCL_i + ETA_{CL})/(TVV_i + ETA_V)^{t_{ij}}}\right)^2}{W_{ij}}$$

$$C_{ij,pred} = C_{ij,pred}(\theta) + \frac{\partial C_{ij,pred}(\theta)}{\partial \eta_i} \times \eta_i + \epsilon_{ij} + \xi_{ij}$$

$$E_{C_{ij,pred}} = C_{ij,pred}(\theta) + \frac{\partial C_{ij,pred}(\theta)}{\partial \eta_i} \times E(\eta) + E(\epsilon) + E(\xi_{ij})$$

$$E_{C_{ij,pred}} = C_{ij,pred}(\theta)$$

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First-Order Approximation

- Using the First-order Taylor Series Expansion
- And set ETAs and EPSILONs to their expected value of zero
- We get an objective function that
  - Estimates the THETAs for all the covariate effects
  - Which provides typical values of PK parameters
- OK, BUT WHAT ABOUT THE OMEGAs???

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We still need to appropriately weight the residuals

$$OFV_{WLS} = \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{W_{ij}}$$

weight was equal to the variance of the residual

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Journal of Pharmacokinetics and Biopharmaceutics, Vol. 5, No. 5, 1977

Estimation of Population Characteristics of Pharmacokinetic Parameters from Routine Clinical Data

Lewis B. Sheiner,[1,4] Barr Rosenberg,[2] and Vinay V. Marathe[2,3]

Received December 3, 1976—Final April 13, 1977

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Eqn 1-6 develop covariate models for TVCL and TVV using 4 THETAs
And add the 2 ETA terms
And a 2-parameter RUV model
And take the required Taylor Series expansion

$$C_{ij} = M_{ij}\left[1 + \left(\frac{Cl_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)\eta_i^V - \frac{t_{ij}}{V_{d_i}}\eta_i^{Cl}\right] + (\alpha^2 M_{ij}^2 + 1)^{1/2}\varepsilon_{ij} \qquad (7)$$

where

$$M_{ij} = \frac{FD_i}{V_{d_i}} \exp\left(-\frac{Cl_i}{V_{d_i}} t_{ij}\right)$$

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mean and its variance. Its mean is M_{ij}. Its variance is the appropriately weighted sum of the variances of its additive random components, and their

456 **Sheiner, Rosenberg, and Marathe**

covariances:

$$\begin{aligned} \mathrm{var}(C_{ij}) = {} & M_{ij}^2\left(\frac{Cl_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)^2 \sigma_V^2 + M_{ij}^2\left(\frac{t_{ij}}{V_{d_i}}\right)^2 \sigma_{Cl}^2 \\ & - 2\left(\frac{Cl_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)\left(\frac{t_{ij}}{V_{d_i}}\right) M_{ij}^2 \sigma_{V,Cl} + (\alpha^2 M_{ij}^2 + 1)\sigma_\varepsilon^2 \end{aligned} \qquad (8)$$

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$$C_{ij} = M_{ij}\left[1 + \left(\frac{\mathrm{Cl}_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)\cancelto{0}{\eta_i} - \frac{t_{ij}}{V_{d_i}}\cancelto{0}{\eta_i}\right] + (\alpha^2 M_{ij}^2 + 1)^{1/2}\cancelto{0}{\varepsilon_{ij}} \quad (7)$$

where

$$M_{ij} = \frac{FD_i}{V_{d_i}} \exp\left(-\frac{\mathrm{Cl}_i}{V_{d_i}} t_{ij}\right)$$

CL_i and Vd_i are functions of all the thetas and covariate effects and can be estimated

covariances:

$$\mathrm{var}(C_{ij}) = M_{ij}^2\left(\frac{\mathrm{Cl}_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)^2 \sigma_V^2 + M_{ij}^2\left(\frac{t_{ij}}{V_{d_i}}\right)^2 \sigma_{\mathrm{Cl}}^2 - 2\left(\frac{\mathrm{Cl}_i}{V_{d_i}} t_{ij} - \frac{1}{V_{d_i}}\right)\left(\frac{t_{ij}}{V_{d_i}}\right) M_{ij}^2 \sigma_{V,\mathrm{Cl}} + (\alpha^2 M_{ij}^2 + 1)\sigma_\varepsilon^2 \quad (8)$$

Random effect variances ω^2 and σ^2 can be estimated

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We get all the population PK parameter estimates,
Along with their variances,
And the magnitudes of all covariate effects,
And the residual variability

All without ever directly estimating the PK in any individual

!!!

!!!

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We still need to weight residuals

$$\mathrm{OFV_{WLS}} = \left[\sum_i \sum_j \frac{(\mathrm{C_{ij,obs}} - \mathrm{C_{ij,pred}})^2}{\mathrm{Var}}\right]$$

The BAD news

$$\mathrm{var}(C_{ij}) = M_{ij}^2\left(\frac{\mathrm{Cl}_i}{V_{d_i}^2} t_{ij} - \frac{1}{V_{d_i}}\right)^2 \sigma_V^2 + M_{ij}^2\left(\frac{t_{ij}}{V_{d_i}}\right)^2 \sigma_{\mathrm{Cl}}^2 - 2\left(\frac{\mathrm{Cl}_i}{V_{d_i}} t_{ij} - \frac{1}{V_{d_i}}\right)\left(\frac{t_{ij}}{V_{d_i}}\right) M_{ij}^2 \sigma_{V,\mathrm{Cl}} + (\alpha^2 M_{ij}^2 + 1)\sigma_\varepsilon^2 \quad (8)$$

The GOOD news

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Objective Function Value Equation Extended Least Squares

$$\mathrm{OFV_{WLS}} = \left[\sum_i \sum_j \frac{(\mathrm{C_{ij,obs}} - \mathrm{C_{ij,pred}})^2}{\mathrm{Var}}\right]$$

$$\mathrm{OFV_{ELS}} = \left[\sum_i \sum_j \frac{(\mathrm{C_{ij,obs}} - \mathrm{C_{ij,pred}})^2}{\mathrm{Var}} + \ln(\mathrm{VAR}_j)\right]$$

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Population vs. Individual

- NONMEM estimates the population parameters (means and variances)
- NOT the individual's parameters
- So how do we get back to the individual?

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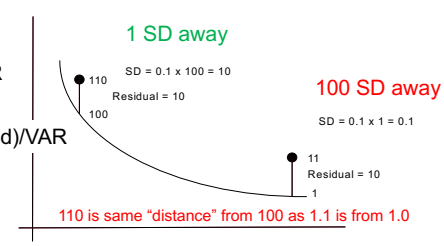
Recall CV=10% (Cobs – Cpred) Residual
Convert to Weighted Residuals (WRES)

WT = 1/VAR

(Cobs – Cpred)/VAR

WRES now in SD units



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Now use the same rationale to think about this as PK parameters instead of concentrations

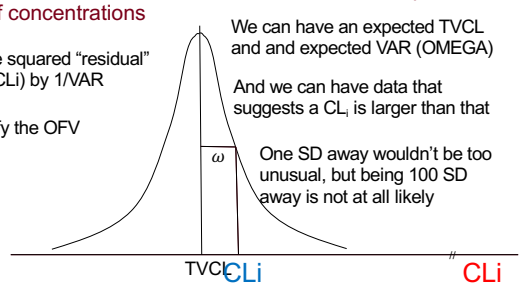
Weight the squared "residual" (CLi – TVCLi) by 1/VAR

And modify the OFV

We can have an expected TVCL and and expected VAR (OMEGA)

And we can have data that suggests a CL_i is larger than that

One SD away wouldn't be too unusual, but being 100 SD away is not at all likely



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Objective Function Value Equation (Empirical) Bayesian Least Squares

$$OFV_{ELS} = \left[\sum_i \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{Var} + \ln(VAR_j)\right]$$

$$OFV_{BLS} = \left[\sum_i \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{Var} + \ln(VAR_j)\right] + \frac{(CLi - TVCLi)^2}{VAR_{CL}} + \frac{(Vi - TVVi)^2}{VAR_v} + \cdots$$

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Maximum a Posteriori (MAP) Bayesian

- AFTER NONMEM estimates population means and variances
- NONMEM uses an individual's data with the estimated means and variances as population priors to get CL_i and V_i

$$OFV_{BLS} = \left[\sum_i \sum_j \frac{(C_{ij,obs} - C_{ij,pred})^2}{Var} + \ln(VAR_j)\right] + \frac{(CL_i - TVCLi)^2}{VAR_{CL}} + \frac{(V_i - TVVi)^2}{VAR_v} + \cdots$$

- NONMEM can calculate the Empirical Bayes Estimates (EBEs) of each subject's parameters
- This can happen with even one data point in a subject (Sparse Data)

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FO

- First Order Estimation Method (FO)
- Uses OFV_{ELS}
- All ETAs are set to zero
- POST HOC command causes NONMEM to run OFV_{BLS} once at the end of the run to get the individual parameter values (EBEs)

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FO vs FOCE

- First Order Conditional Estimation Method (FOCE)
- Method FOCE causes NONMEM to use OFV_{BLS} at **EACH ITERATION** to get conditional ETAs
  - ETAs are not set to zero
  - ETA_is are used in the OFV
  - Allows the parameter space to be searched differently
- Causes **MUCH** longer run times
- FOCE is preferred
- FO is mostly of historical interest

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PRED vs. IPRED

- PRED is the prediction based on the population-based regression equations

  TVCL = THETA(1)+THETA(2)*WT+…..
- All the ETAs are set to zero

  C_{pred} = D/TVV * exp[-(TVCL/TVV)*t]
- IPRED is the prediction based on CL_i, not TVCL, with individual ETAs calculated

  $CL_i = TVCL_i + ETA(1)$ $V_i = TVV_i + ETA(2)$

  $C_{ipred} = D/V_i * \exp[-(CL_i/V_i)*t]$

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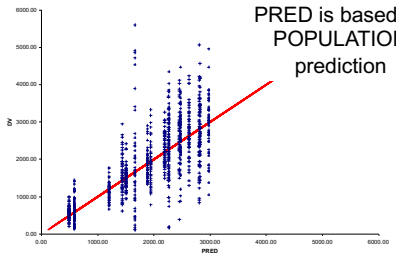

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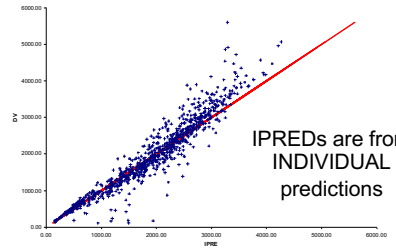

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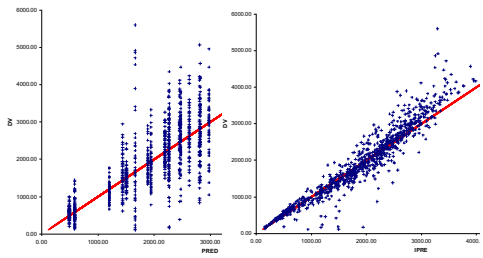

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WRES vs CWRES

- WRES are calculated using the PRED, and then weighted
- **EVEN WHEN** using FOCE
- It's a historical definition that's been carried forward
  - (Original NONMEM did not have FOCE, only FO)
- CWRES are calculated with the RES based on IPREDs, and then weighted
- You will usually see CWRES in plots rather than WRES

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Probability to Likelihood to ELS

Assume a model *f* with parameters θ

$$y_i = f(x_i, \theta) + \varepsilon_i$$

Then, assuming ε_i is ~ N(0; σ_i^2)

$$p(y_i|f(\theta)) = \frac{1}{\sqrt{2\pi\sigma_i^2}} exp^{-\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}}$$

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But we have a (large) collection of *N* observations

The joint probability is the product of the individual probabilities

$$p(data|f(\theta)) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_i^2}} exp^{-\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}}$$

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$$\mathcal{L}(data|f(\theta)) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_i^2}} \quad exp^{-\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}}$$

$$ln(\mathcal{L}) = \sum_{i=1}^{N} ln\left(\frac{1}{\sqrt{2\pi\sigma_i^2}} \quad exp^{-\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}}\right)$$

$$ln(\mathcal{L}) = \sum_{i=1}^{N}\left[\ln\left(2\pi\sigma_i^2\right)^{-\frac{1}{2}} + ln\left(exp^{-\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}}\right)\right]$$

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$$ln(\mathcal{L}) = \sum_{i=1}^{N}\left[\ln\left(2\pi\sigma_i^2\right)^{-\frac{1}{2}} - \left(\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}\right)\right]$$

$$ln(\mathcal{L}) = -\sum_{i=1}^{N}\left[\frac{1}{2}ln(2\pi) + \frac{1}{2}ln(\sigma_i^2) \quad + \left(\frac{[y_i - f(x_i;\theta)]^2}{2\sigma_i^2}\right)\right]$$

$$-2ln(\mathcal{L}) = \sum_{i=1}^{N}\left[ln(2\pi) + \ln(\sigma_i^2) \quad + \left(\frac{[y_i - f(x_i;\theta)]^2}{\sigma_i^2}\right)\right]$$

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$$-2ln(\mathcal{L}) \propto \sum_{i=1}^{N}\left(\frac{[y_i - f(x_i;\theta)]^2}{\sigma_i^2}\right) + ln(\sigma^2)$$

$$\mathrm{OFV_{ELS}} = \sum_{\mathrm{j}} \frac{\left(\mathrm{C_{ij,obs}} - \mathrm{C_{ij,pred}}\right)^2}{\mathrm{VAR}_j} + \ln(\mathrm{VAR}_j)$$

Maximizing the likelihood of the data is equivalent to minimizing the ELS OFV!!!

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